International Baccalaureate®
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MARKSCHEME

SPECIMEN

CHEMISTRY

Standard Level

Paper 2

10 pages

Subject Details: Chemistry SL Paper 2 Markscheme

Mark Allocation

Candidates are required to answer **ALL** questions. Maximum total = **[50 marks]**.

1. Each row in the "Question" column relates to the smallest subpart of the question.

2. The maximum mark for each question subpart is indicated in the "Total" column.

3. Each marking point in the "Answers" column is shown by means of a tick (✓) at the end of the marking point.

4. A question subpart may have more marking points than the total allows. This will be indicated by "**max**" written after the mark in the "Total" column. The related rubric, if necessary, will be outlined in the "Notes" column.

5. An alternative wording is indicated in the "Answers" column by a slash (/). Either wording can be accepted.

6. An alternative answer is indicated in the "Answers" column by "***OR***" on the line between the alternatives. Either answer can be accepted.

7. Words in angled brackets ‹ › in the "Answers" column are not necessary to gain the mark.

8. Words that are <u>underlined</u> are essential for the mark.

9. The order of marking points does not have to be as in the "Answers" column, unless stated otherwise in the "Notes" column.

10. If the candidate's answer has the same "meaning" or can be clearly interpreted as being of equivalent significance, detail and validity as that in the "Answers" column then award the mark. Where this point is considered to be particularly relevant in a question it is emphasized by ***OWTTE*** (or words to that effect) in the "Notes" column.

11. Remember that many candidates are writing in a second language. Effective communication is more important than grammatical accuracy.

12. Occasionally, a part of a question may require an answer that is required for subsequent marking points. If an error is made in the first marking point then it should be penalized. However, if the incorrect answer is used correctly in subsequent marking points then **follow through** marks should be awarded. When marking, indicate this by adding **ECF** (error carried forward) on the script. "ECF acceptable" will be displayed in the "Notes" column.

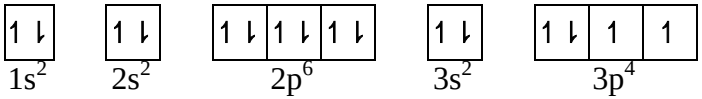
13. Do **not** penalize candidates for errors in units or significant figures, **unless** it is specifically referred to in the "Notes" column.

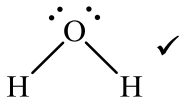
14. If a question specifically asks for the name of a substance, do not award a mark for a correct formula unless directed otherwise in the "Notes" column, similarly, if the formula is specifically asked for, unless directed otherwise in the "Notes" column do not award a mark for a correct name.

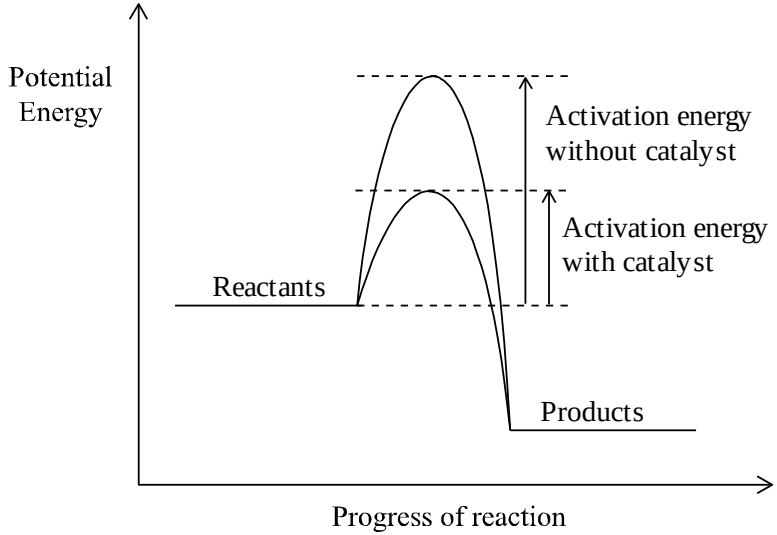
15. If a question asks for an equation for a reaction, a balanced symbol equation is usually expected, do not award a mark for a word equation or an unbalanced equation unless directed otherwise in the "Notes" column.

16. Ignore missing or incorrect state symbols in an equation unless directed otherwise in the "Notes" column.

Question			Answers	Notes	Total
1.	**a**	**i**	‹$(22.05+22.15)(0.5)$ =›22.10‹cm^3› ✓		**1**
	a	**ii**	$\left\langle \frac{22.10\times0.100}{1000} = \right\rangle 2.21\times10^{-3}$ / 0.00221‹mol› ✓		**1**
	a	**iii**	$\left\langle \frac{0.5\times2.21\times10^{-3}\times1000}{25.00} = \right\rangle 4.42\times10^{-2}$ / 0.0442‹$mol\,dm^{-3}$› ✓		**1**
	a	**iv**	‹$4.42\times10^{-2}\times10$ =›4.42×10^{-1} / 0.442‹$mol\,dm^{-3}$› ✓		**1**
	b	**i**	*NaClO*: +1 ‹for chlorine› **and** I_2: 0 ‹for iodine› ✓		**1**
	b	**ii**	ClO^- since chlorine reduced/gains electrons ***OR*** ClO^- since oxidation state of chlorine changes from +1 to –1/decreases ***OR*** ClO^- since it loses oxygen / causes iodide to be oxidized ✓		**1**
	b	**iii**	produces chlorine ‹gas›/Cl_2 ‹on reaction with ClO^-› which is toxic ✓	*OWTTE*	**1**
	b	**iv**	oxidation states are not real ***OR*** oxidation states are just used for electron book-keeping purposes ✓ average oxidation state of sulfur calculated to be +2 ✓ but the two sulfurs are bonded differently/in different environments in thiosulfate so have different oxidation states ✓	*OWTTE*	**2 max**

Question			Answers	Notes	Total
	c		*Valid:* addition of oxygen signifies an oxidation reaction so C is oxidized ***OR*** loss of hydrogen signifies an oxidation reaction so C is oxidized ***OR*** oxidation state of C changes from –4 to +4/increases ✓ *Not valid:* loss of electrons might suggest formation of ionic product but not valid since CO_2 is covalent ***OR*** loss of electrons might suggest formation of ionic product but not valid since reaction only involves neutral molecules ✓	*OWTTE*	**2**
	d	**i**	$[Ne]3s^2 3p^4$ ✓	*Electrons must be given as superscript.*	**1**
	d	**ii**	 ✓		**1**

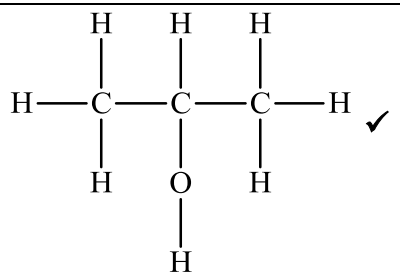
Question			Answers	Notes	Total
2.	**a**	**i**	radical / unpaired electron ✓		**1**
	a	**ii**	<table><tr><th>Molecule</th><th>Lewis (electron dot) structure</th></tr><tr><td>SO_2</td><td> ✓</td></tr><tr><td>H_2O</td><td>  ✓</td></tr></table>	*Lines, x's or dots may be used to represent electron pairs.*	**2**
	a	**iii**	<table><tr><th></th><th>Electron domain geometry</th><th>Molecular geometry</th></tr><tr><td>SO_2</td><td>trigonal/triangular planar</td><td>bent/v-shaped/angular✓</td></tr><tr><td>H_2O</td><td>tetrahedral</td><td>bent/v-shaped/angular✓</td></tr></table>	*Award **[1 max]** for either both electron domain geometries correct OR for either both molecular geometries correct.*	**2**
	a	**iv**	*SO_2: Accept any angle in the range greater than 115° but less than 120°.* **and** *H_2O:* 104.5° ✓	*Experimental value is 119°*	**1**

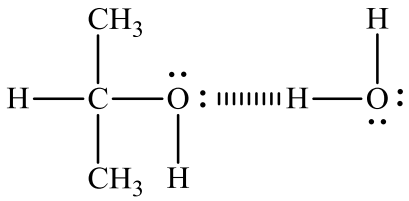
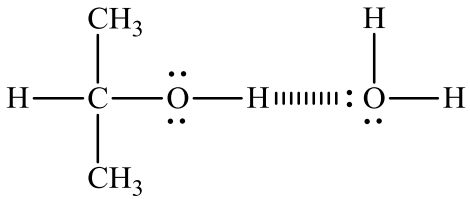
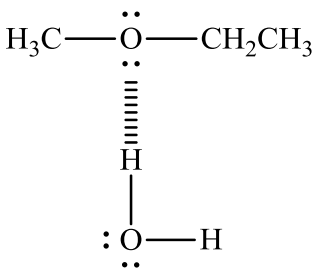
<table>
<tr><th colspan="3">Question</th><th>Answers</th><th>Notes</th><th>Total</th></tr>
<tr><td>2</td><td>a</td><td>v</td><td>
<table>
<tr><th>Change</th><th>Shift</th><th>Reason</th></tr>
<tr><td>Increase in temperature</td><td>LHS</td><td>since ‹forward› exothermic reaction/ $\Delta H < 0$ ✓</td></tr>
<tr><td>Increase in pressure</td><td>RHS</td><td>since fewer ‹gaseous› molecules on RHS ✓</td></tr>
<tr><td>Addition of a catalyst to the mixture</td><td>No change</td><td>since affects rate of forward and reverse reactions equally ✓</td></tr>
</table>
</td><td></td><td>3</td></tr>
<tr><td></td><td>a</td><td>vi</td><td>


correct positions of reactants and products ✓
correct profile with labels showing activation energy with and without a catalyst ✓
</td><td></td><td>2</td></tr>
<tr><td></td><td>a</td><td>vii</td><td>proton/H^+ donor ✓</td><td></td><td>1</td></tr>
<tr><td></td><td>a</td><td>viii</td><td>$HSO_4^-(aq)$ and $H_2O(l)$ ✓</td><td></td><td>1</td></tr>
<tr><td></td><td>a</td><td>ix</td><td>$2NO_2(g) + H_2O(l) \rightarrow HNO_3(aq) + HNO_2(aq)$ ✓</td><td>Ignore state symbols.</td><td>1</td></tr>
</table>

Question			Answers	Notes	Total
3.	**a**		only water/H_2O produced ‹so non-polluting› ✓		**1**
	b		$\Delta H = [(-393.5)] - [(2)(-241.8)]$ ✓ +90.1‹kJ› ✓	*Award **[2]** for correct final answer.*	**2**

4.	**a**		*I:* carboxamide ✓ *II:* phenyl ✓ *III:* carboxyl / carboxy ✓ *IV:* hydroxyl ✓	*Award **[2]** for all four correct, **[1]** for two or three correct.* *Do not allow benzene.* *Do not allow carboxylic/alkanoic acid.* *Do not allow alcohol or hydroxide.*	**2 max**
	b	**i**	$n_C : \left\langle \frac{73.99}{12.01} \right\rangle = 6.161\,(\text{mol})$ **and** $n_H : \left\langle \frac{6.55}{1.01} \right\rangle = 6.49\,(\text{mol})$ **and** $n_N : \left\langle \frac{9.09}{14.01} \right\rangle = 0.649\,(\text{mol})$ **and** $n_O : \left\langle \frac{10.37}{16.00} \right\rangle = 0.6481\,(\text{mol})$ ✓ $n_C : n_H : n_N : n_O = 9.5 : 10 : 1 : 1$ ✓ *Empirical formula*: $C_{19}H_{20}N_2O_2$ ✓	*Award **[2 max]** for correct final answer without working.*	**3**
	b	**ii**	$C_{19}H_{20}N_2O_2$ ✓		**1**
	b	**iii**	$\langle (0.5)(40-20-2) = \rangle 9$ ✓		**1**
	b	**iv**	*A:* C–H **and** *B:* C=O		**1**
	b	**v**	O–H **and** N–H ✓ frequencies/stretches due to O–H and N–H occur above 3200 ‹cm^{-1}› which are not present in IR of *bute* ✓		**2**
	c	**i**	1:1:6 ✓		**1**

Question			Answers	Notes	Total
	c	**ii**	 ✓		**1**
	c	**iii**	$CH_3OCH_2CH_3$ ✓		**1**
	c	**iv**	*Similarity:* both have fragment corresponding to $(M_r - 15)^+$ / both have $m/z = 45$ ✓ *Difference:* **X** has fragment corresponding to $(M_r - 17)^+$ / **X** has $m/z = 43$ ***OR*** **X** has fragment corresponding to $(M_r - 43)^+$ / **X** has $m/z = 17$ ***OR*** **Y** has fragment corresponding to $(M_r - 31)^+$ / **Y** has $m/z = 29$ ***OR*** **Y** has fragment corresponding to $(M_r - 29)^+$ / **Y** has $m/z = 31$ ✓	*Allow "both have same molecular ion peak/M^+ / both have m/z = 60". However in practice the molecular ion peak is of low abundance and difficult to observe for propan-2-ol.*	**2**

Question			Answers	Notes	Total
4.	**c**	**v**	both **X** and **Y** will exhibit hydrogen bonding with water molecules ✓ diagrams showing hydrogen bonding ✓ X:  **OR**  Y: 		**2**
	c	**vi**	*I:* CH_3COCH_3 ✓ *II:* $CH_3COOCH(CH_3)_2$ ✓		**2**
	c	**vii**	propanone ✓		**1**